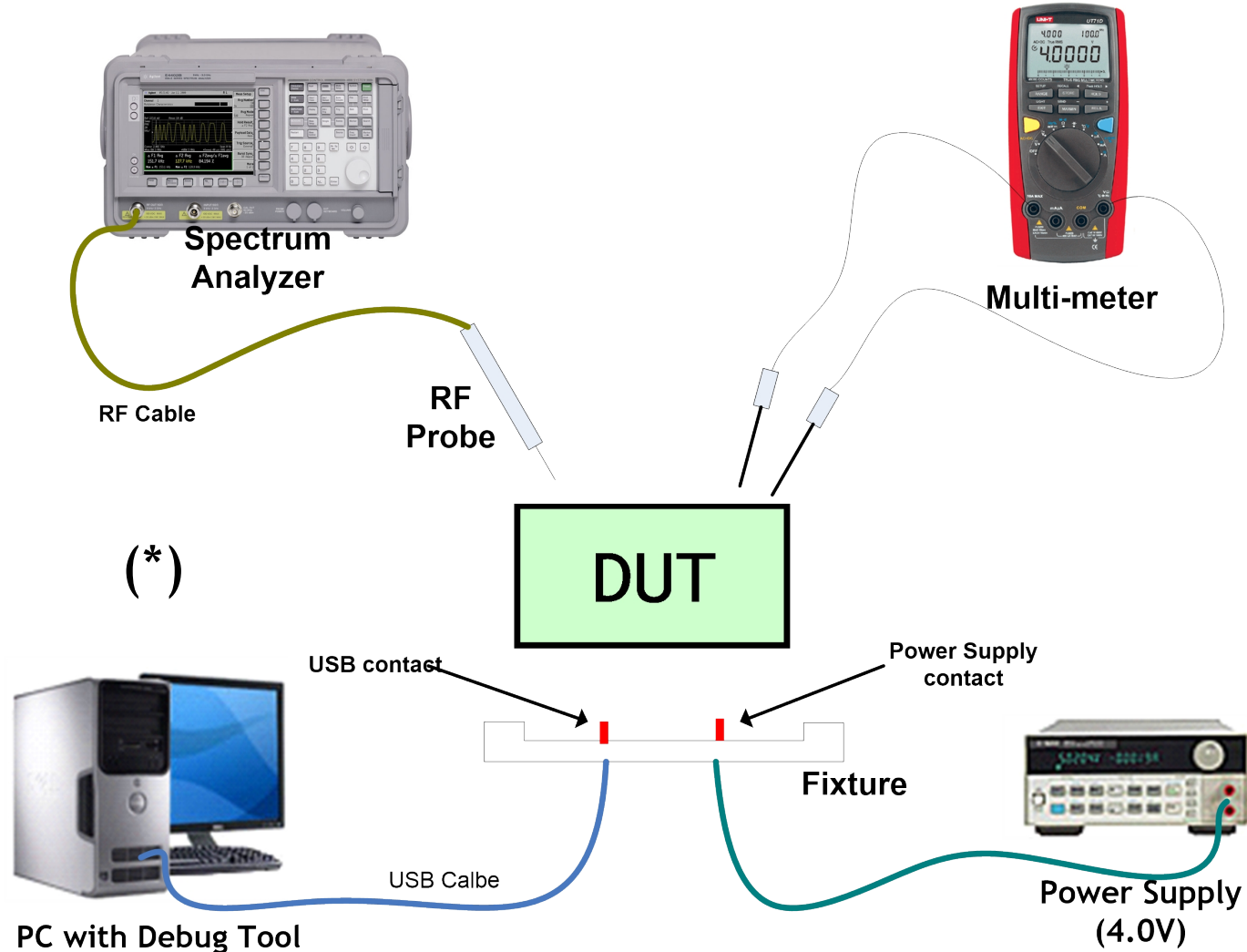


ASUS ZB690KG

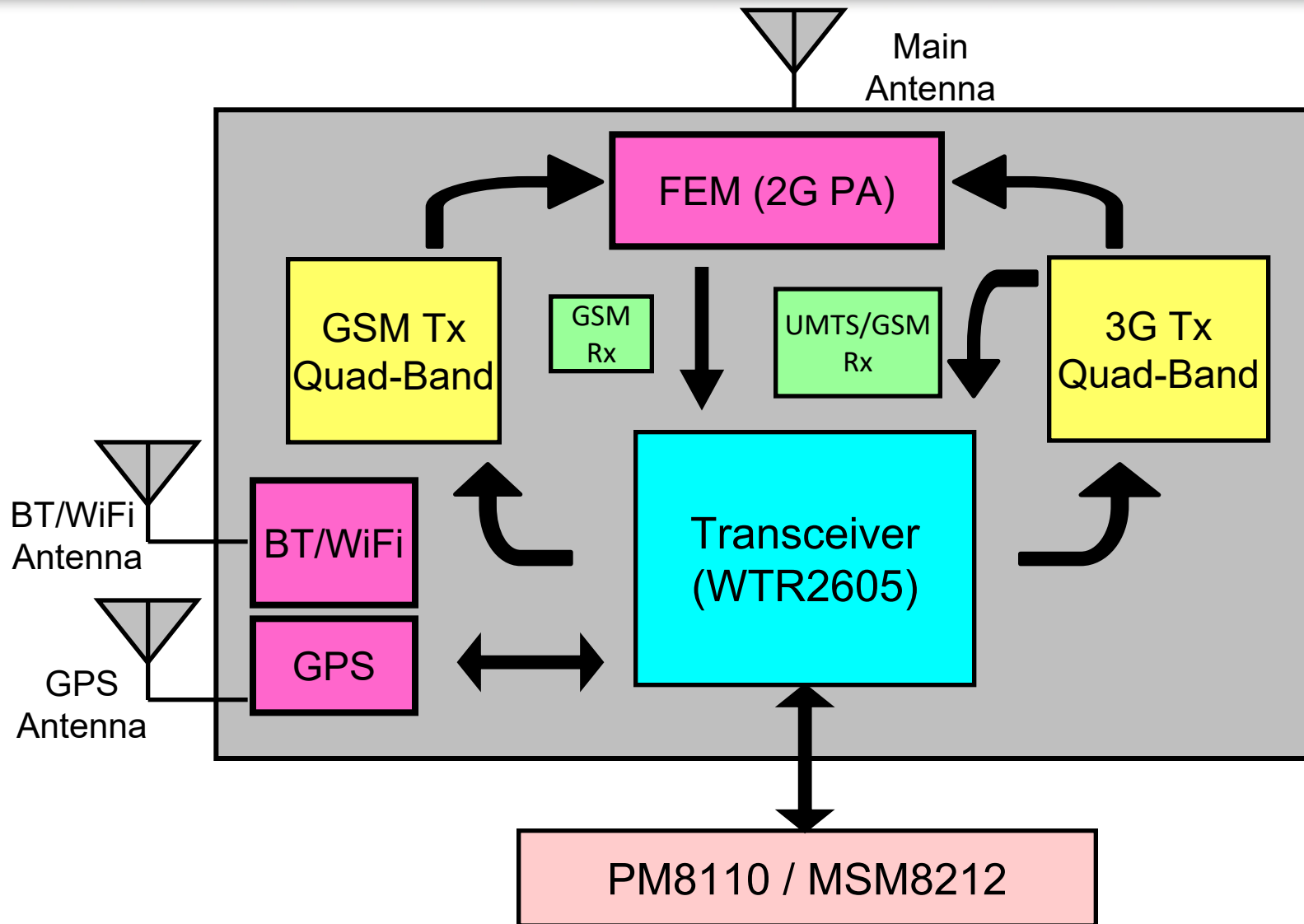
RF Hardware Trouble Shooting Guide

Fred Lee

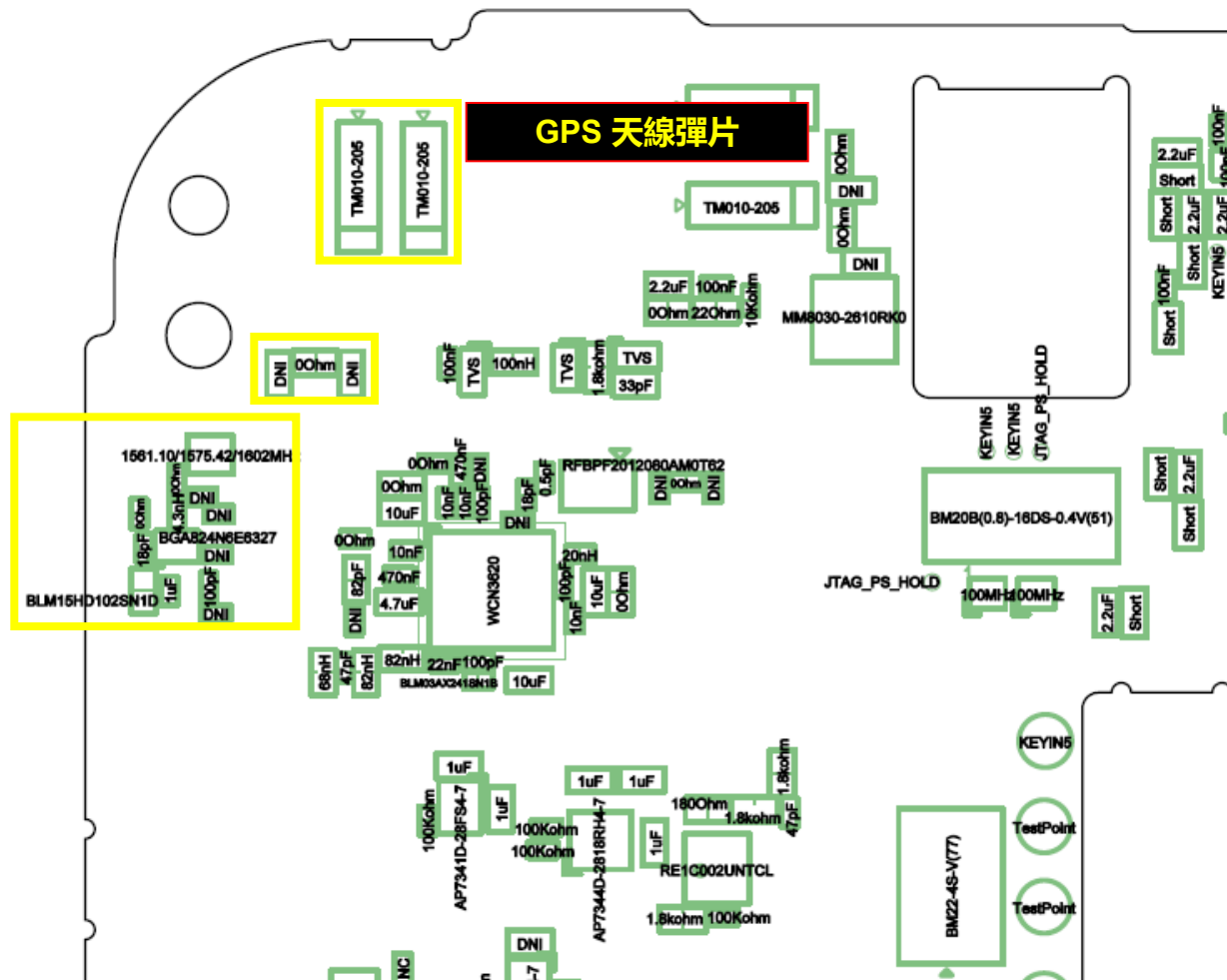
Repair and Test Environment



RF Block Diagram



RF Layout on Bottom



RF Layout on Bottom



RF Layout on Bottom

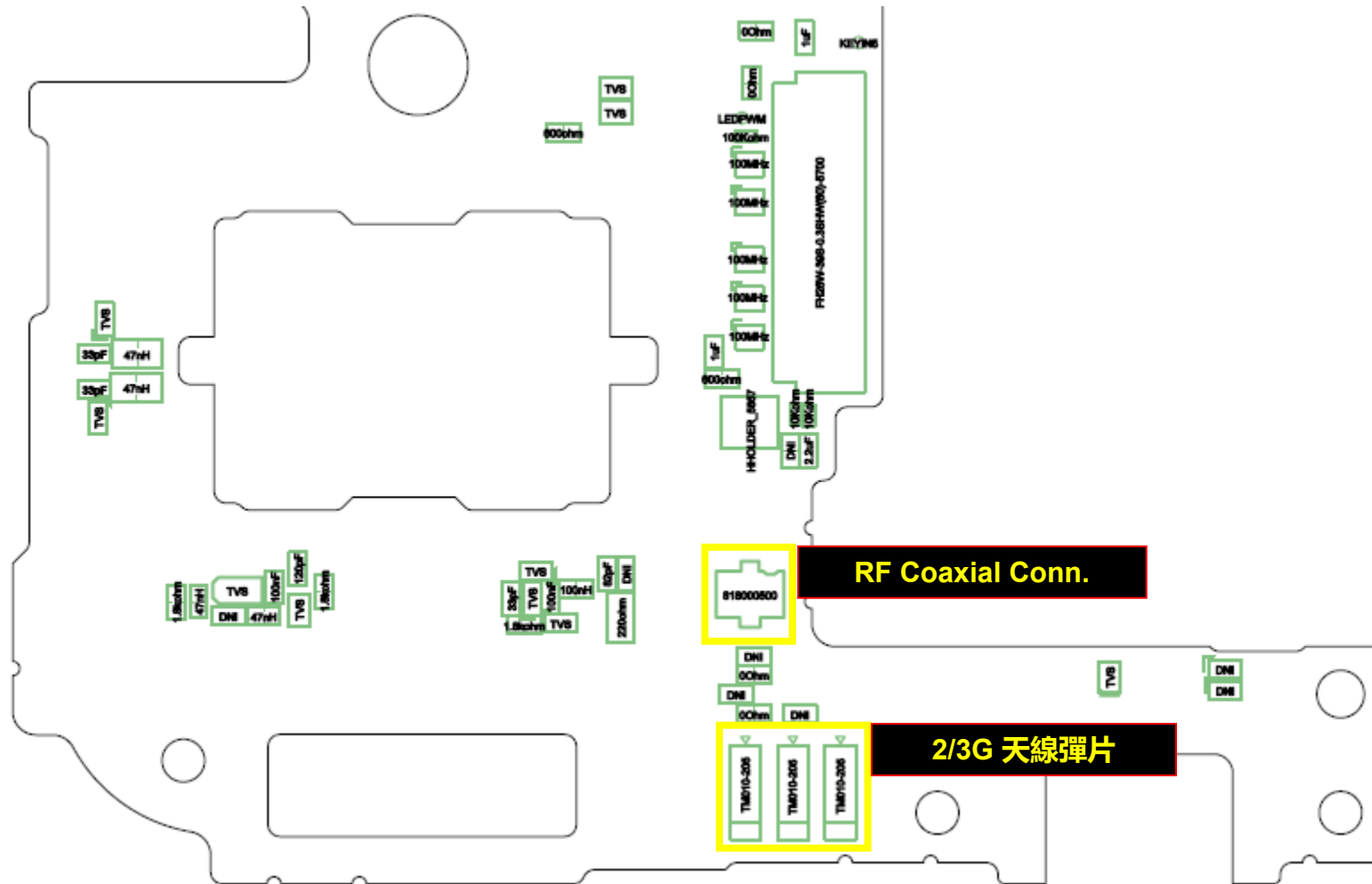


RF Layout on Top

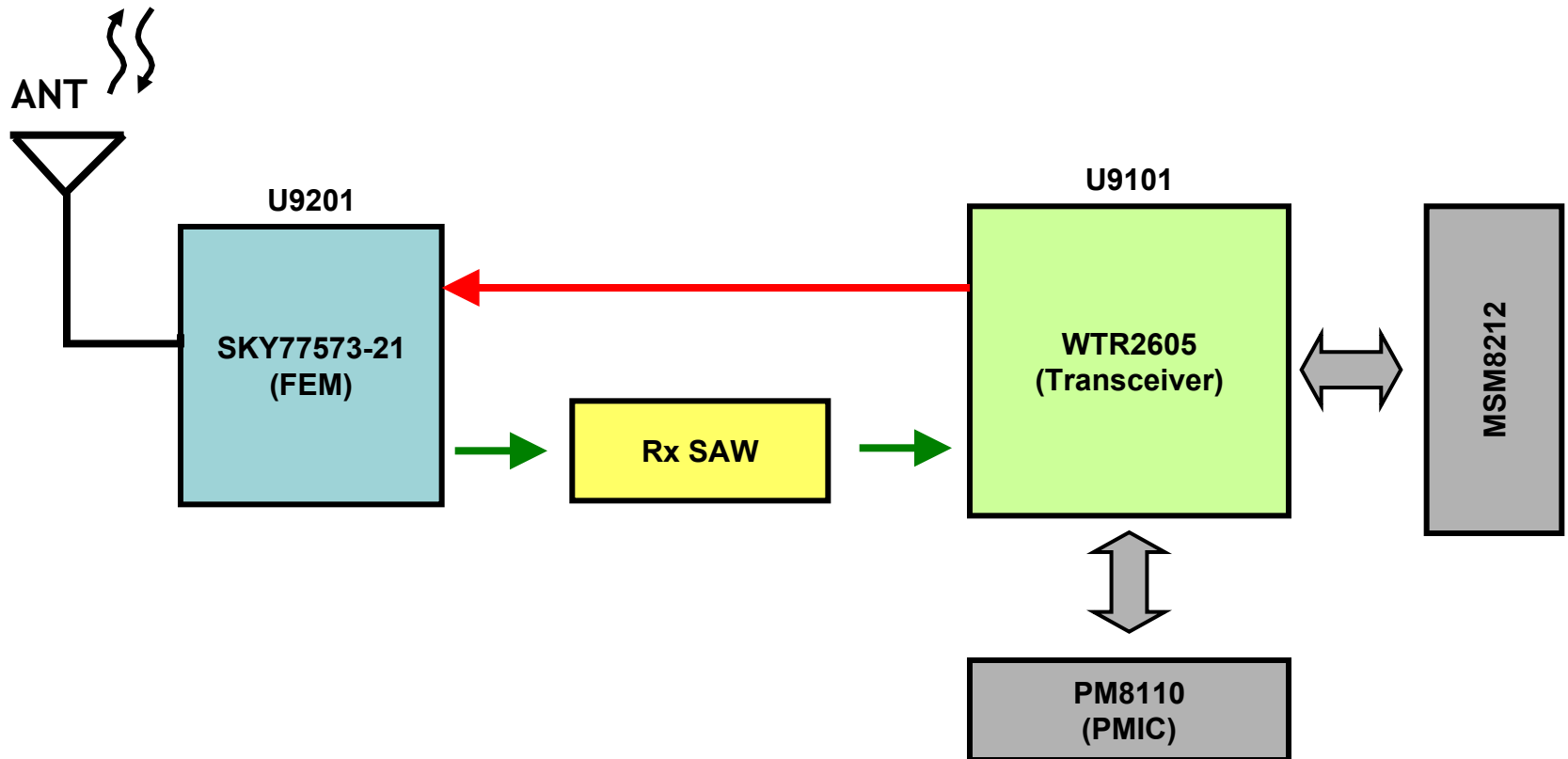


RF areas - GSM/WCDMA

RF Layout on Bottom

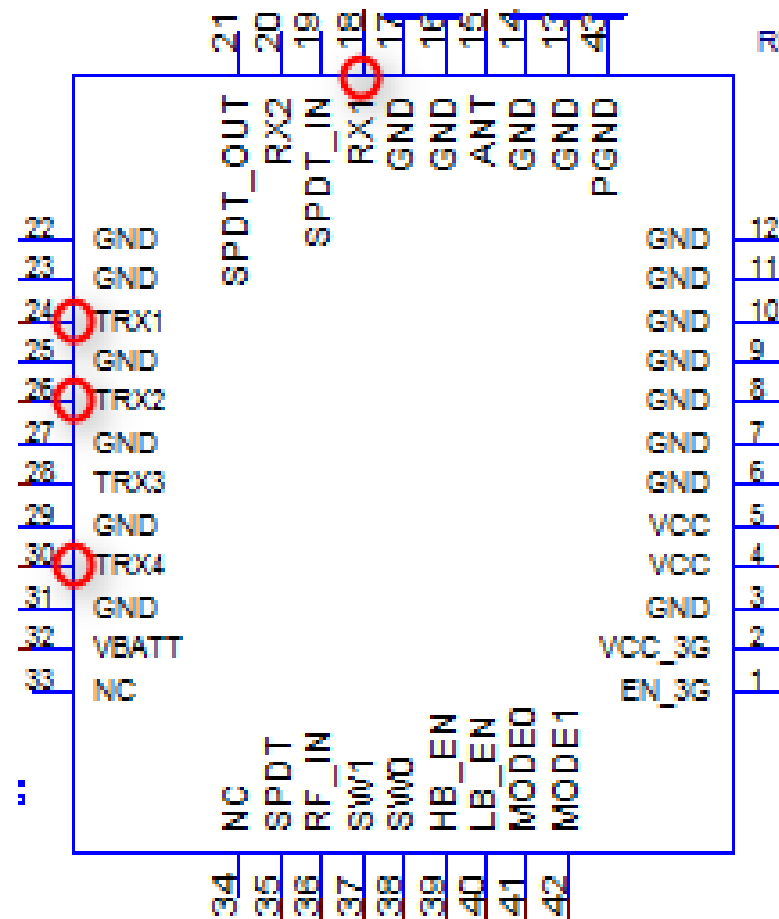


GSM Block Diagram



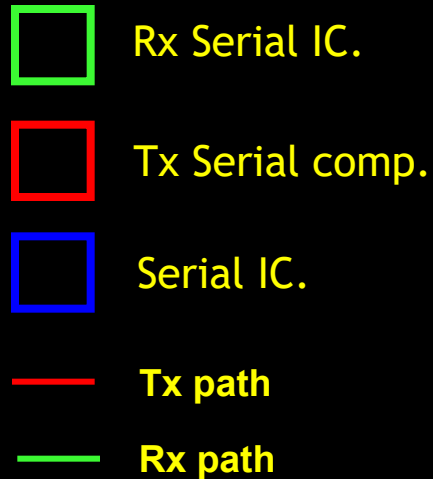
Front End Module(U9101) – SKY77573-21

Antenna Switch SP10T Logic(SKY77573-21)						
ANT PIN	PATH	SPDT	SW1	SW0	HB_EN	LB_EN
TRX1_ANT	TRX_B8_EGSM	1	1	0	1	0
TRX2_ANT	TRX_B5_850	1	1	0	1	1
TRX3_ANT	TRX_B1	1	1	1	0	0
TRX4_ANT	TRX_B2_PCS	1	1	1	0	1
RX1	RF_DCS	1	1	1	1	0
RX2	X	1	1	1	1	1



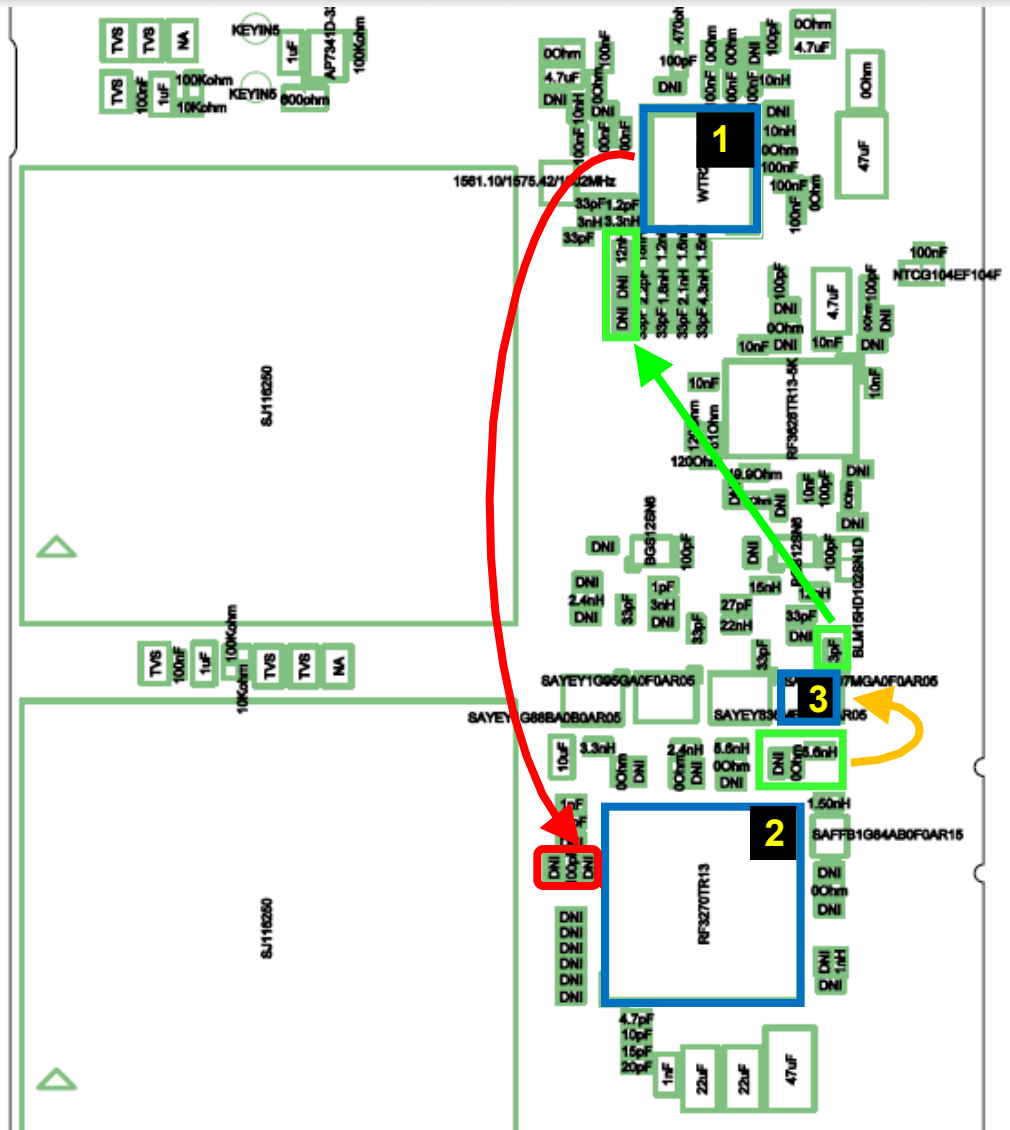
2G System : EGSM900

RF Layout on Bottom



1. U9101: Transceiver
WTR2605
2. U9201: Front End Module
SKY77573-21
3. U9206: Band8 Duplexer
SAYEY897MGA0F0AR0

5

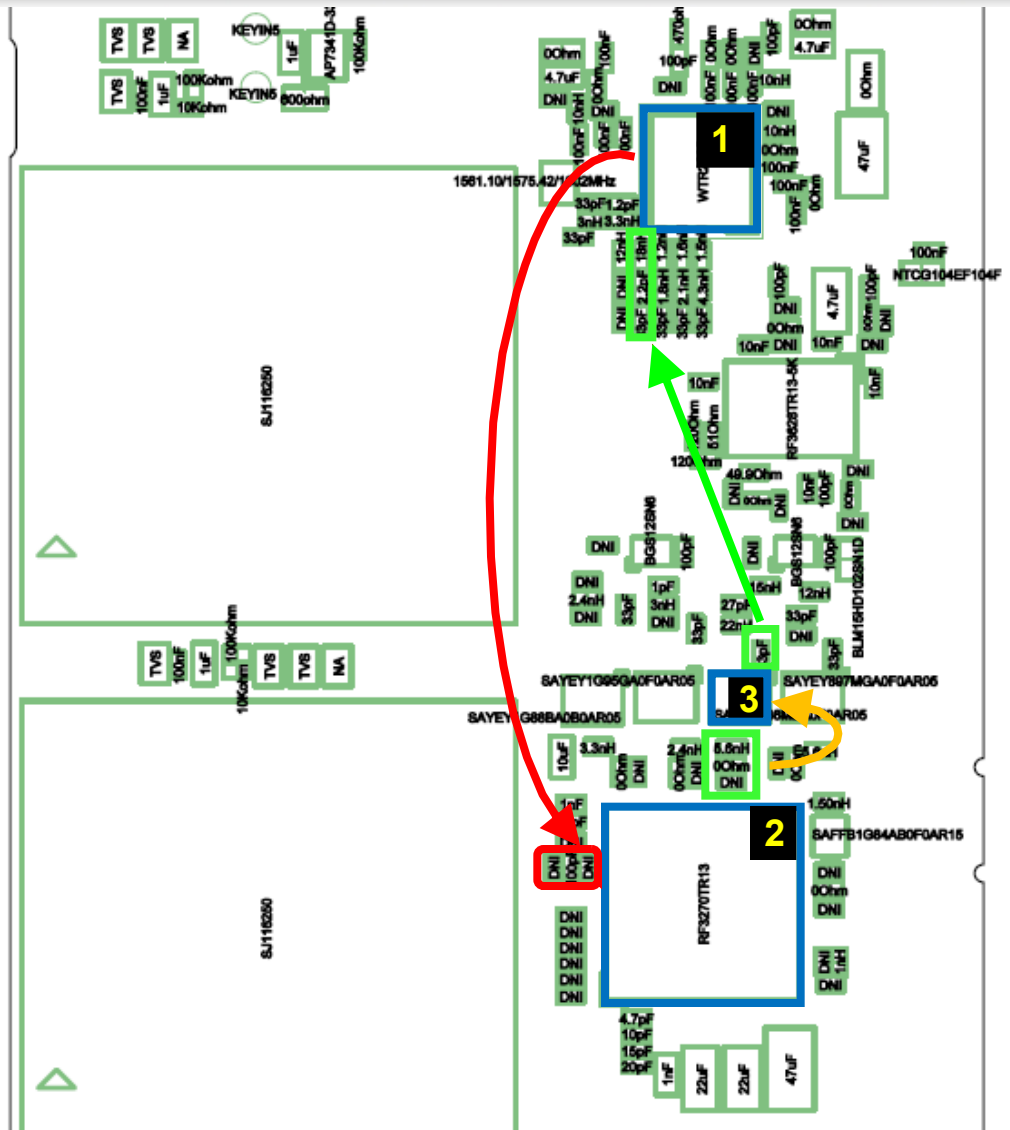


2G System : GSM850

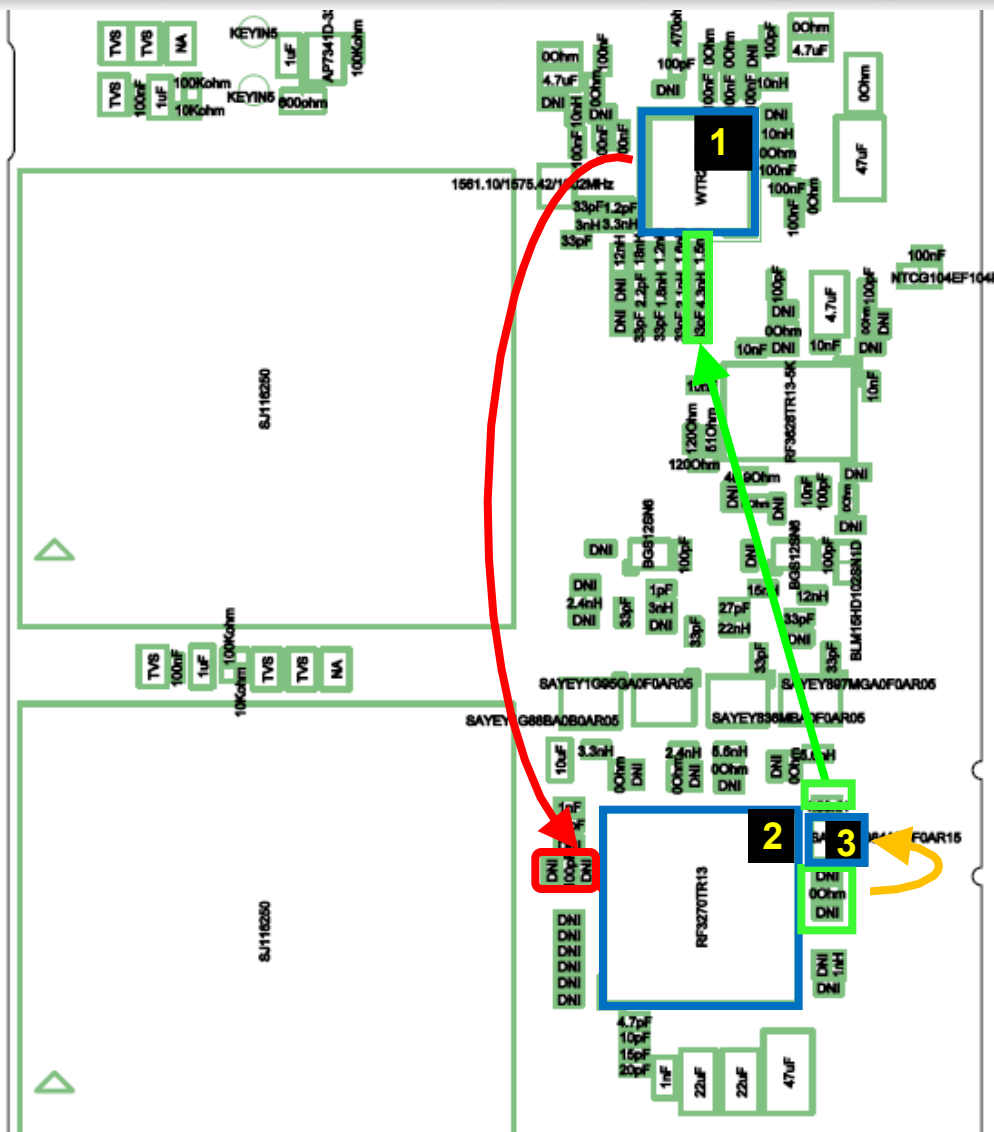
RF Layout on Bottom

-  Rx Serial IC.
-  Tx Serial comp.
-  Serial IC.
-  Tx path
-  Rx path

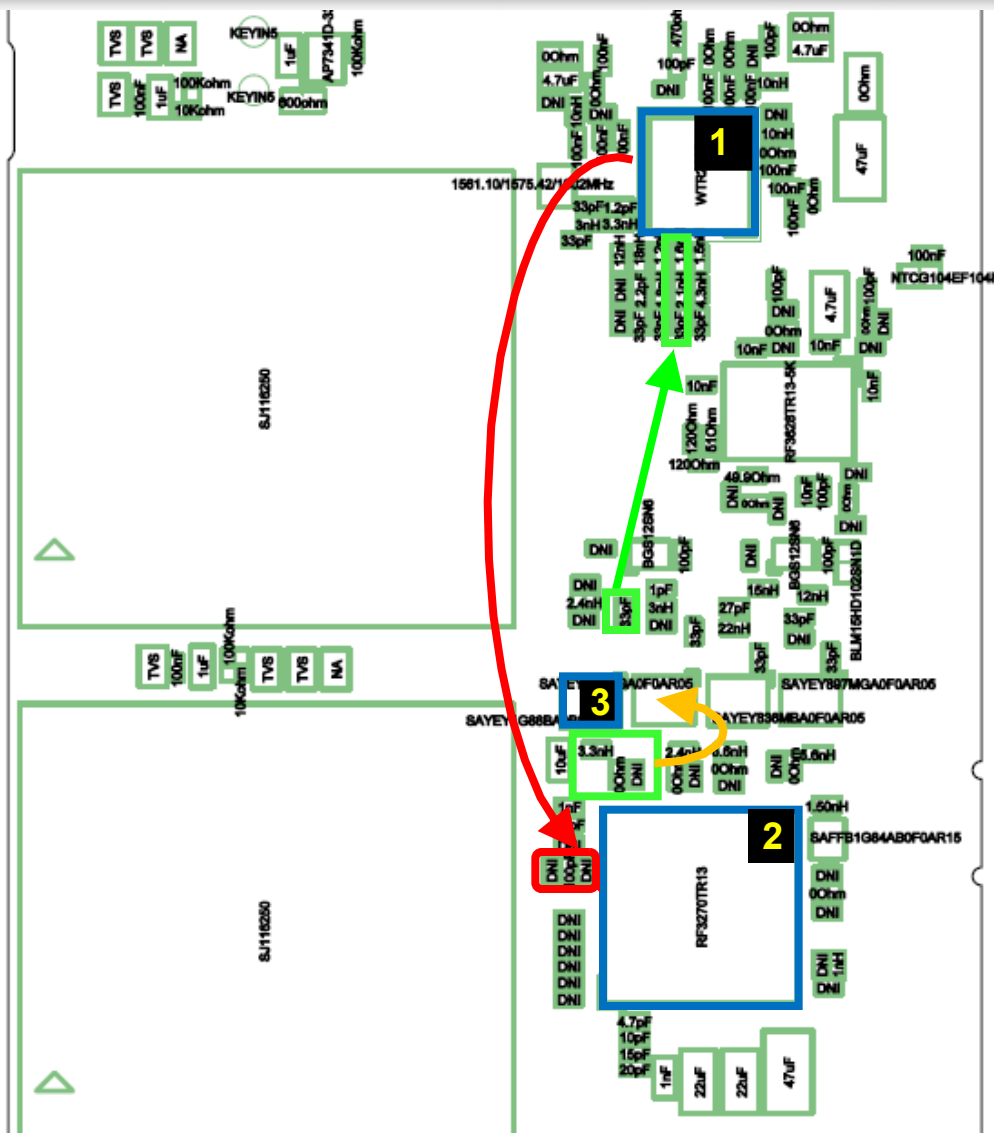
1. U9101: Transceiver
WTR2605
2. U9201: Front End Module
SKY77573-21
3. U9207: Band5 Duplexer
SAYEY836MBA0F0AR05



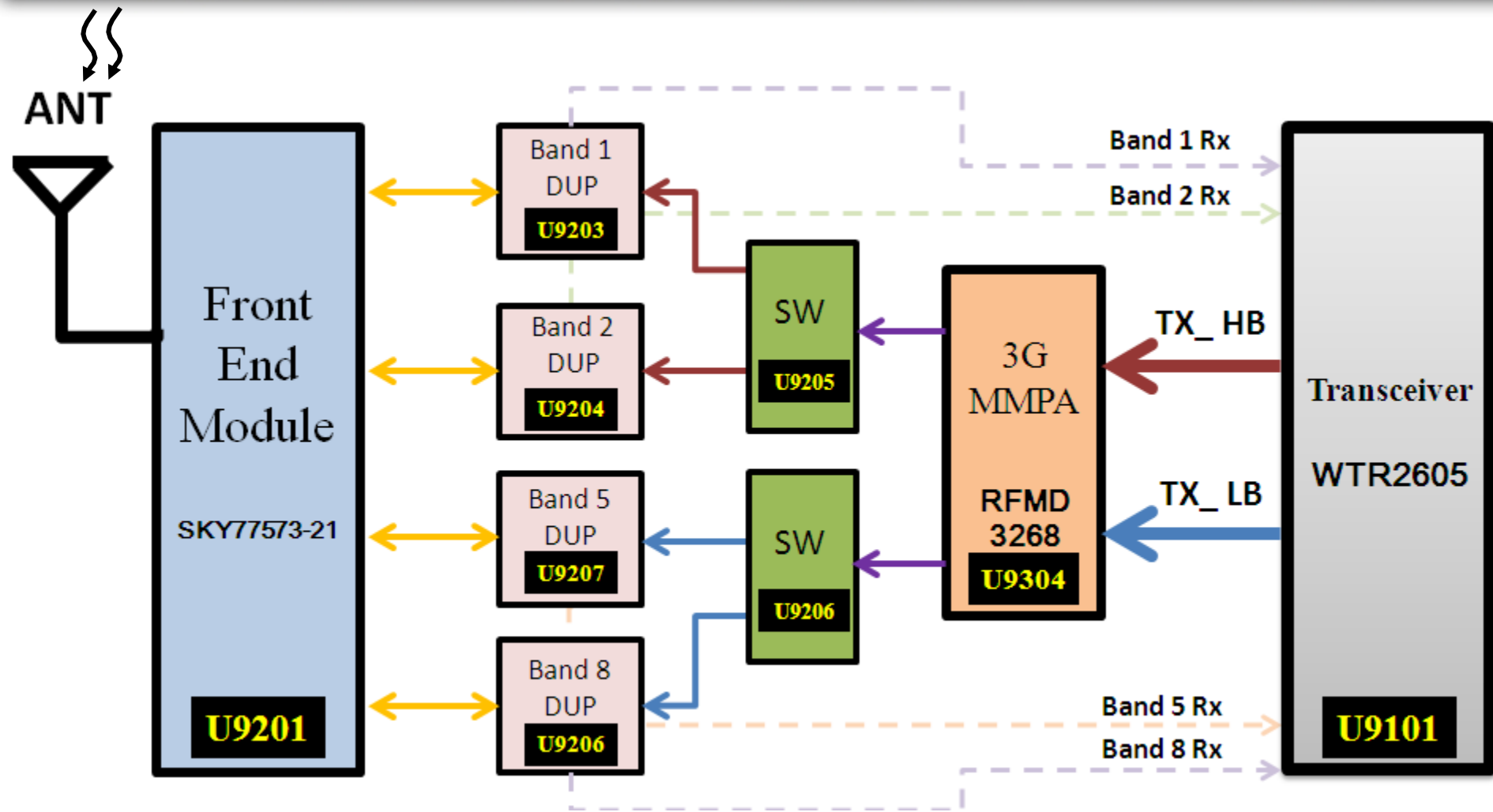
RF Layout on Bottom



RF Layout on Bottom



WCDMA Block Diagram



RF Layout on Bottom



-

RF Layout on Bottom



-

WCDMA Band 5

RF Layout on Bottom

Serial IC.

Rx comp.

Rx path

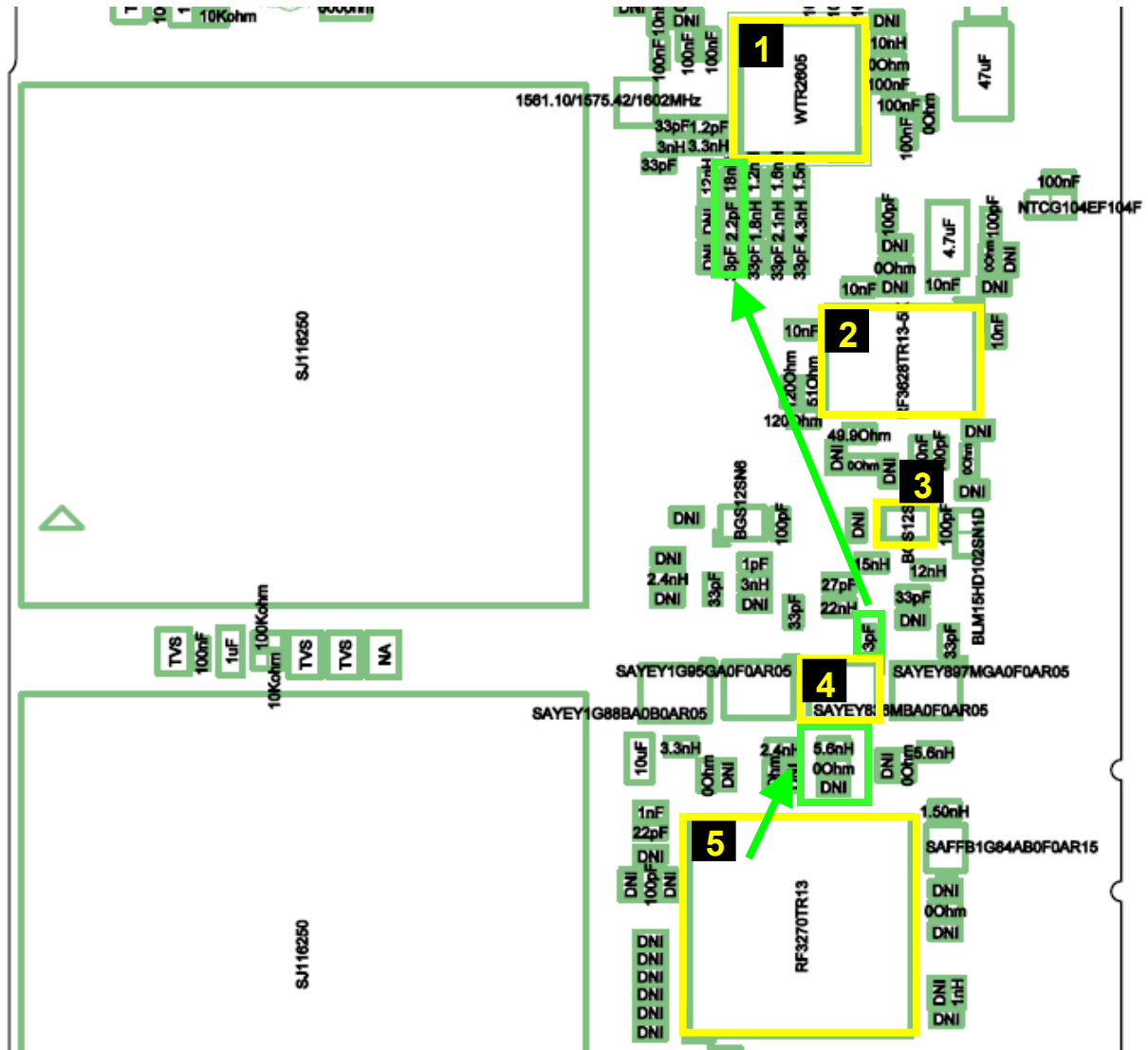
1. U9101: Transceiver
WTR2605

2. U9304: 3G MMPA
RFMD3628

3. U9306 : 3G LB Switch
BGS12SN6

4. U9207: Band5 Duplexer
SAYEY836MBA0F0AR05

5. U9201: Front End Module
SKY77573-21



WCDMA Band 8

RF Layout on Bottom

Serial IC.

Rx comp.

Rx path

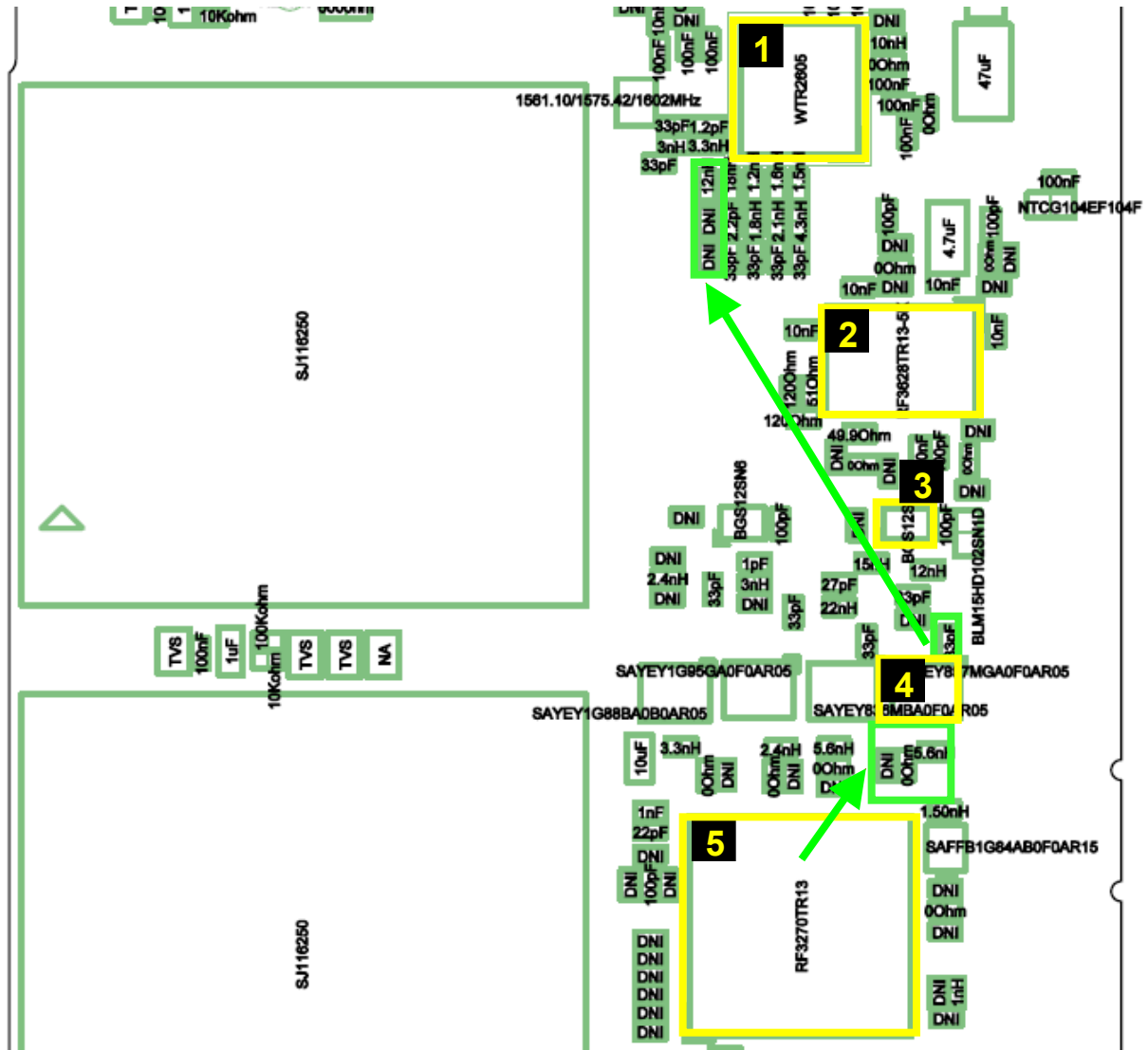
1. U9101: Transceiver
WTR2605

2. U9304: 3G MMPA
RFMD3628

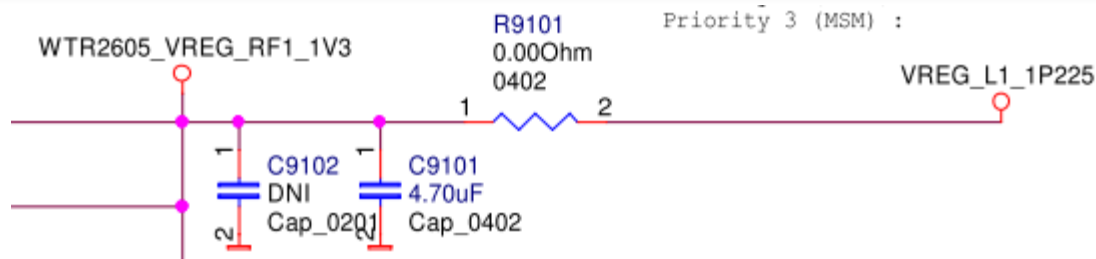
3. U9306 : 3G LB Switch
BGS12SN6

4. U9206: Band8 Duplexer
SAYEY897MGA0F0AR05

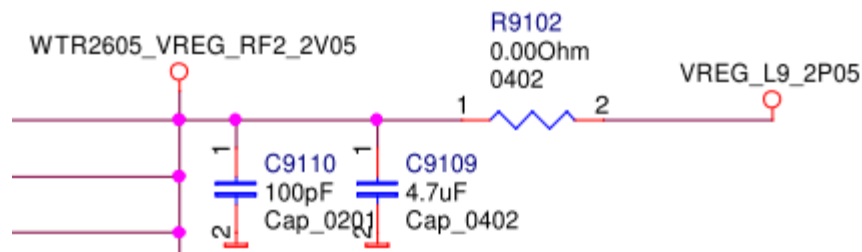
5. U9201: Front End Module
SKY77573-21



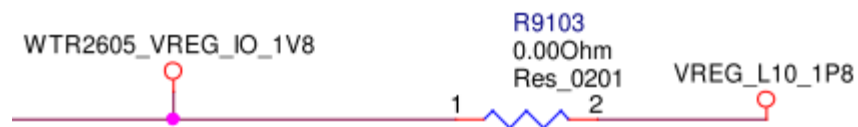
Voltage Setup



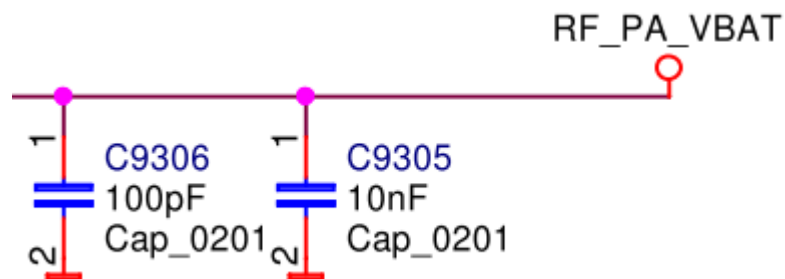
→ 1.225 V



→ 2.05 V



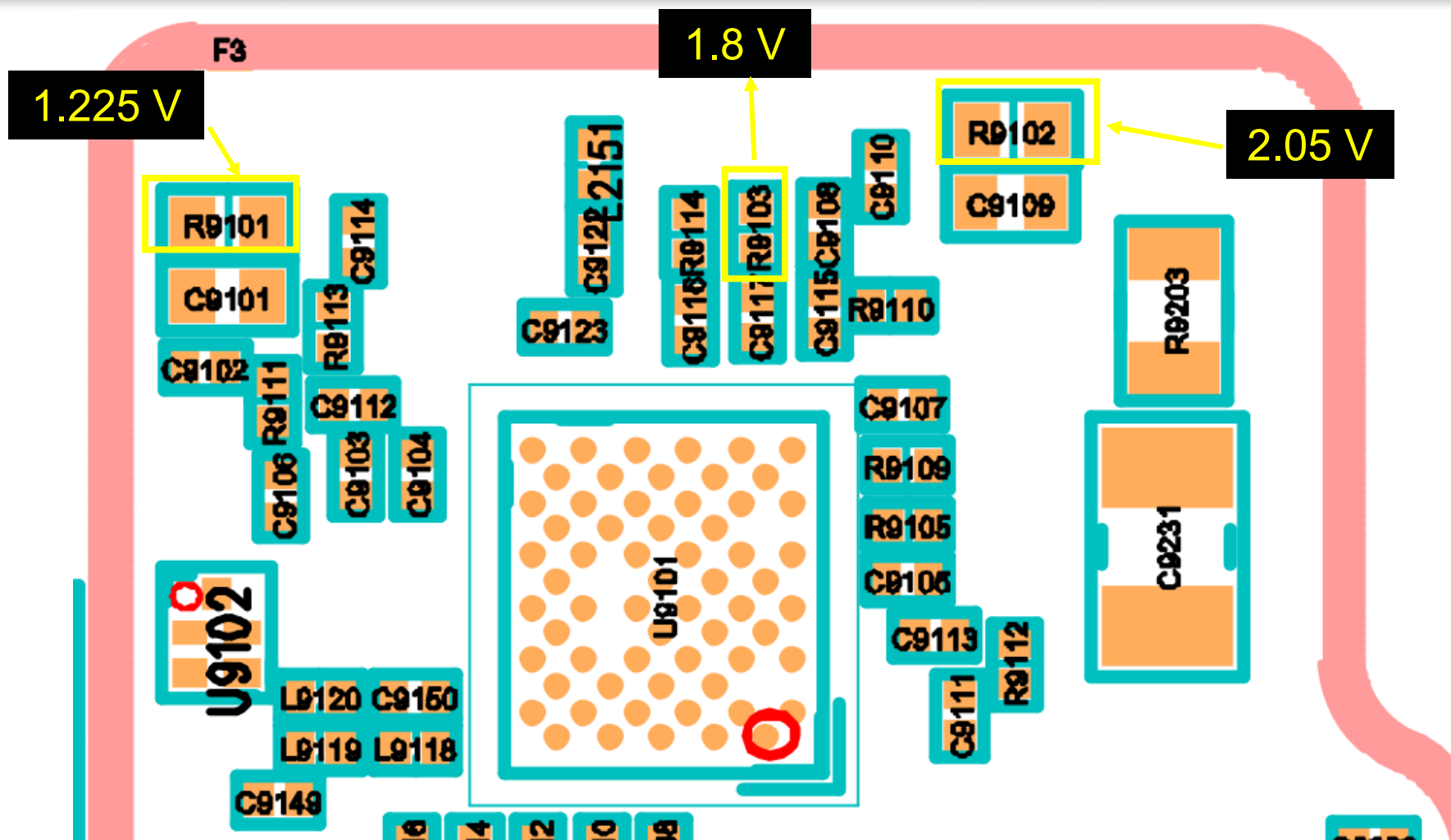
→ 1.8 V



→ 3.8 V

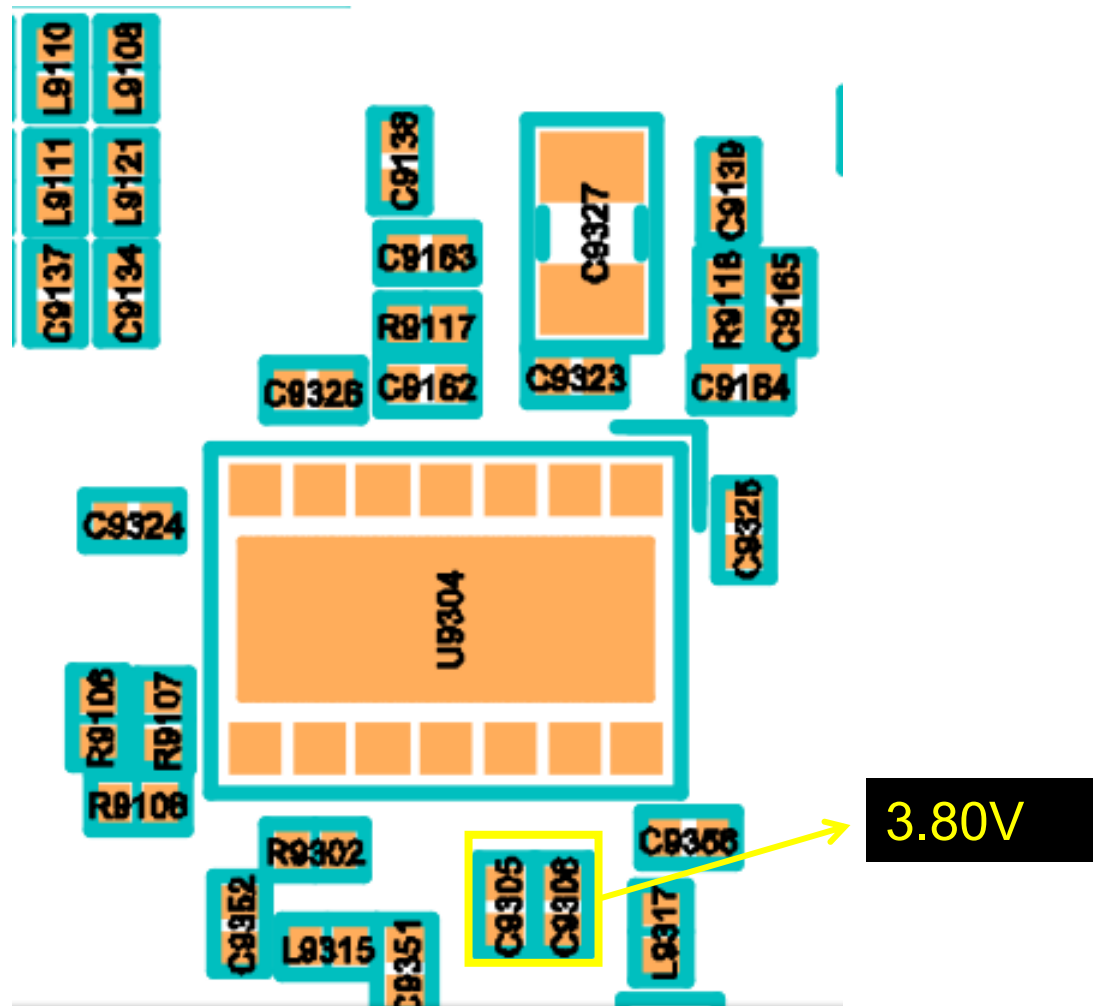
Voltage Setup

RF Layout on Bottom



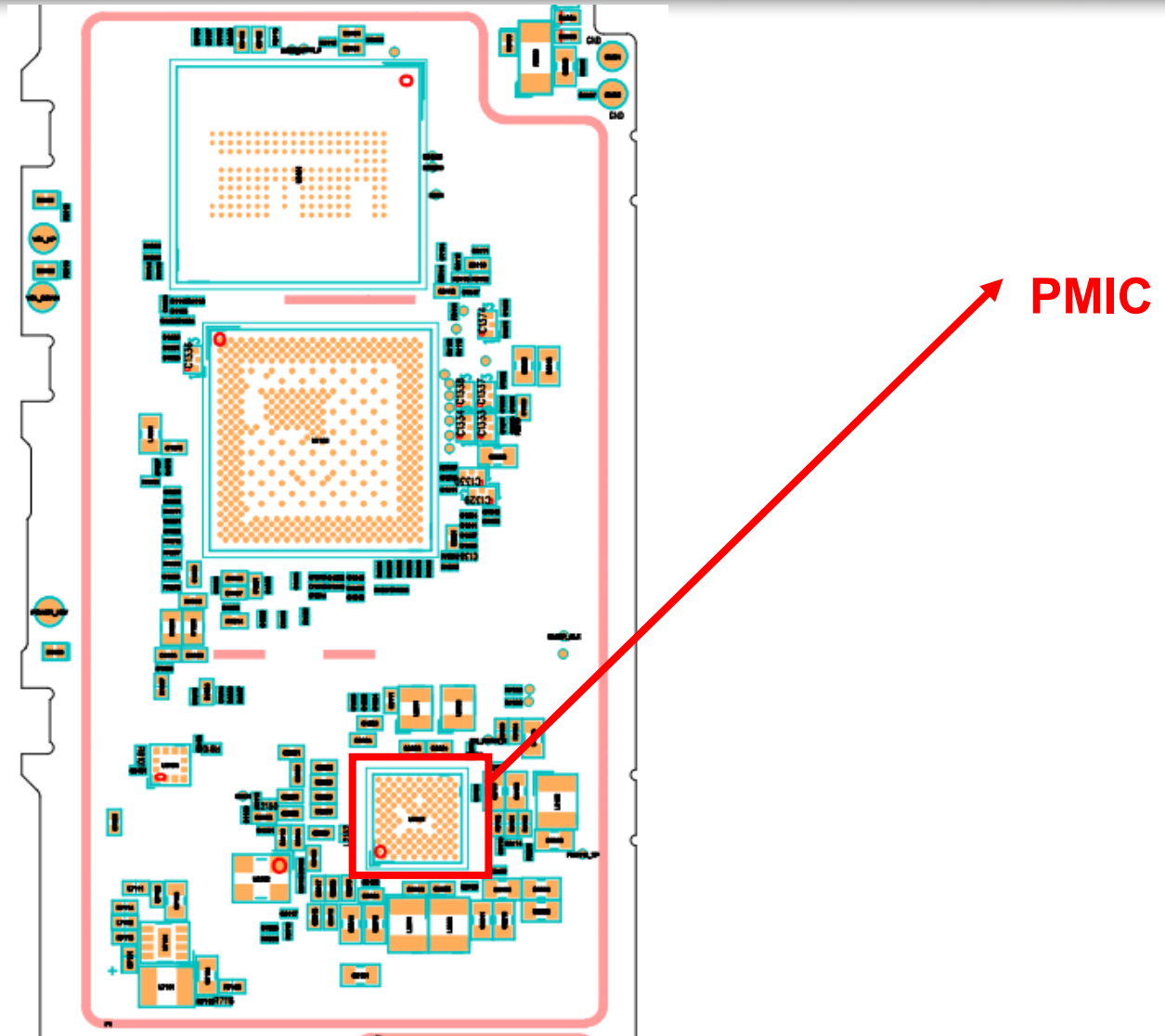
Voltage Setup

RF Layout on Bottom

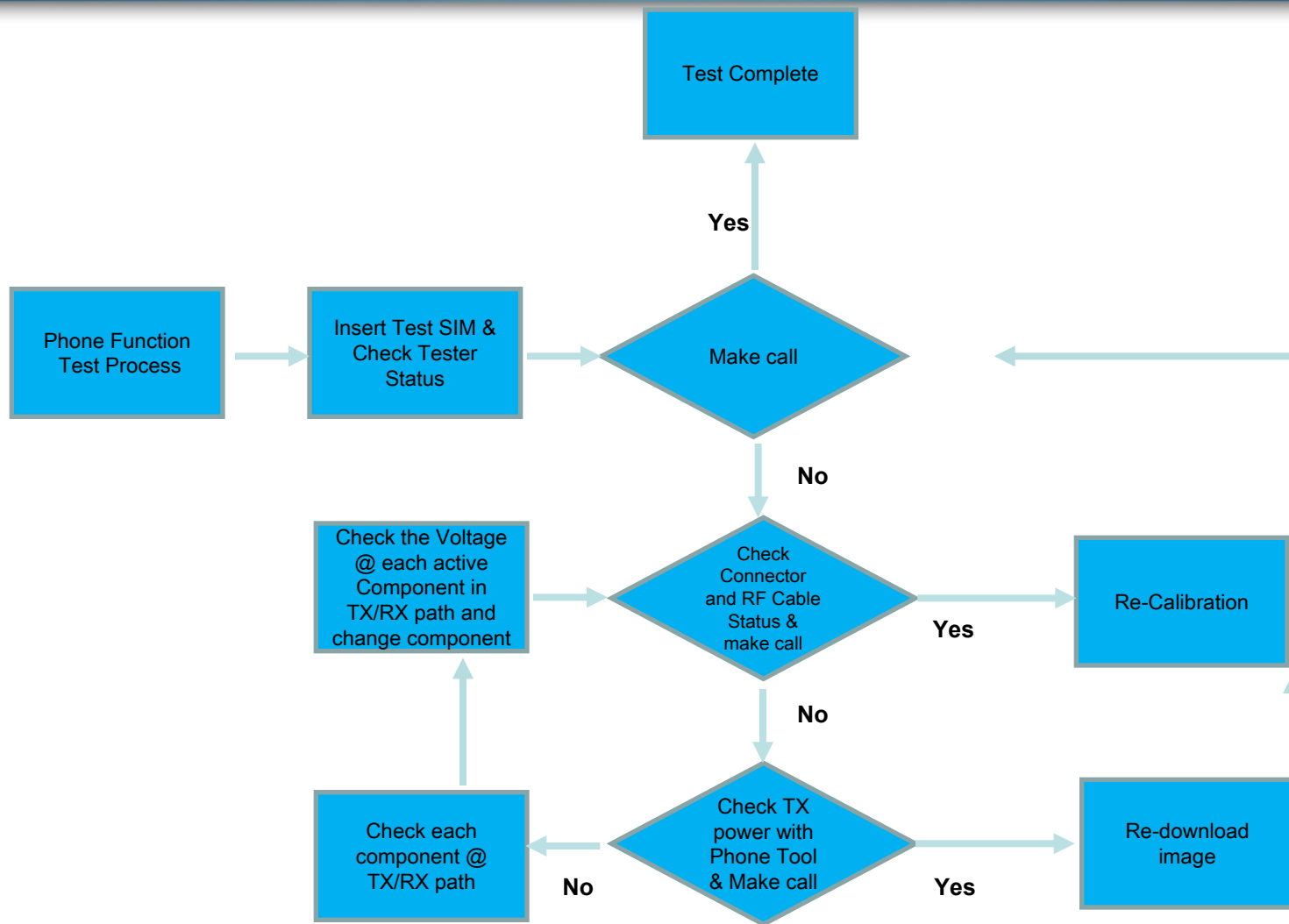


PMIC Location

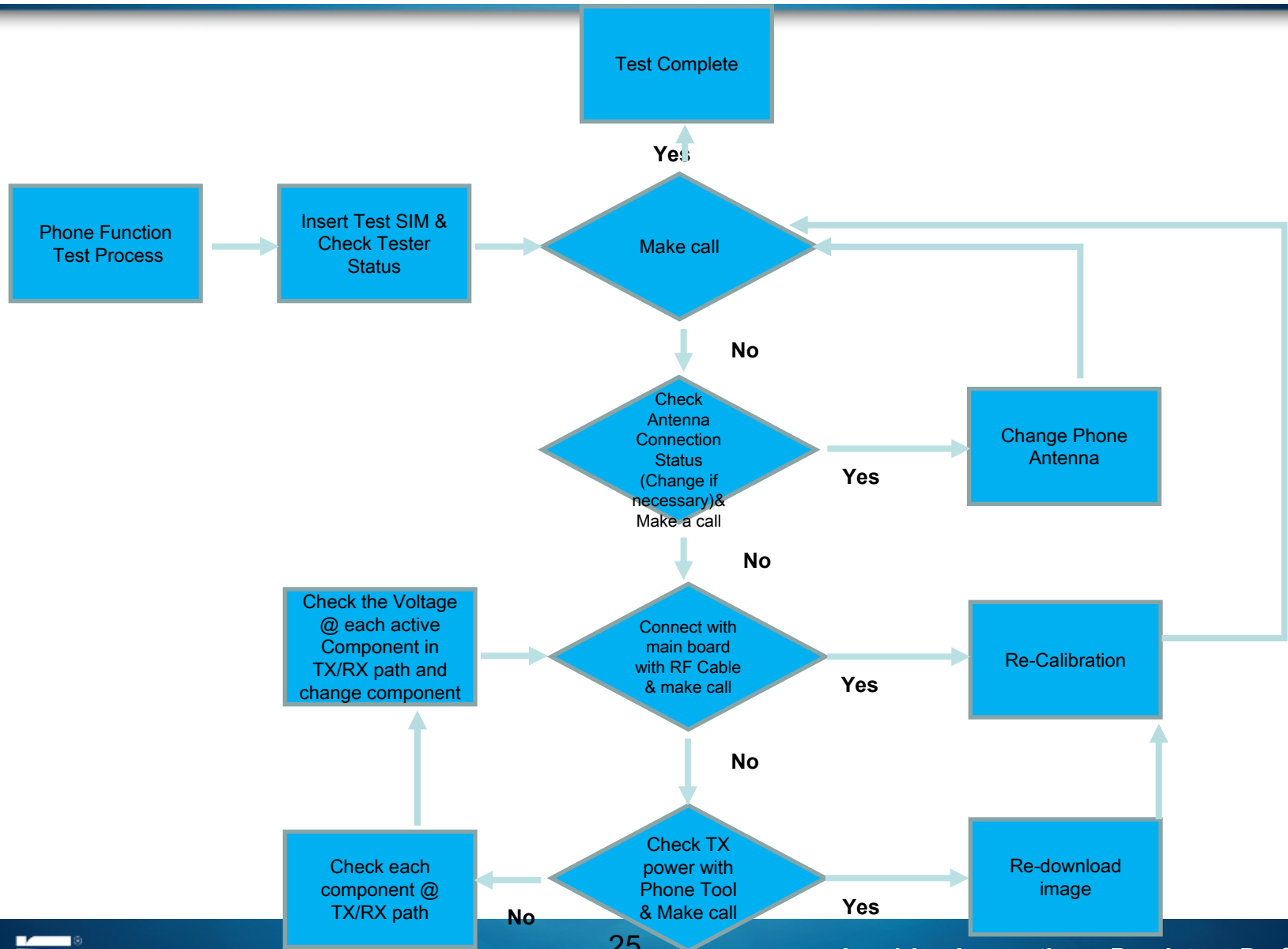
RF Layout on Top



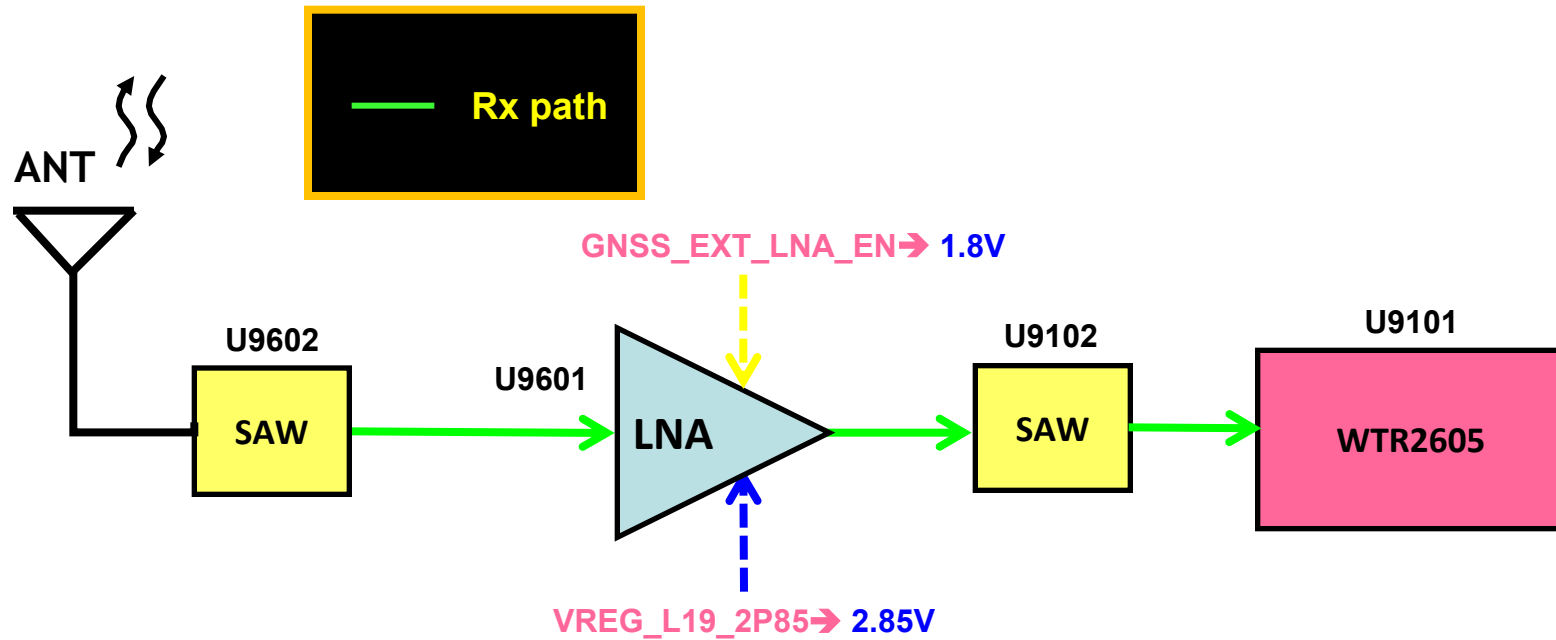
Board Level Phone Function Test Procedure



System Level Phone Function Test Procedure

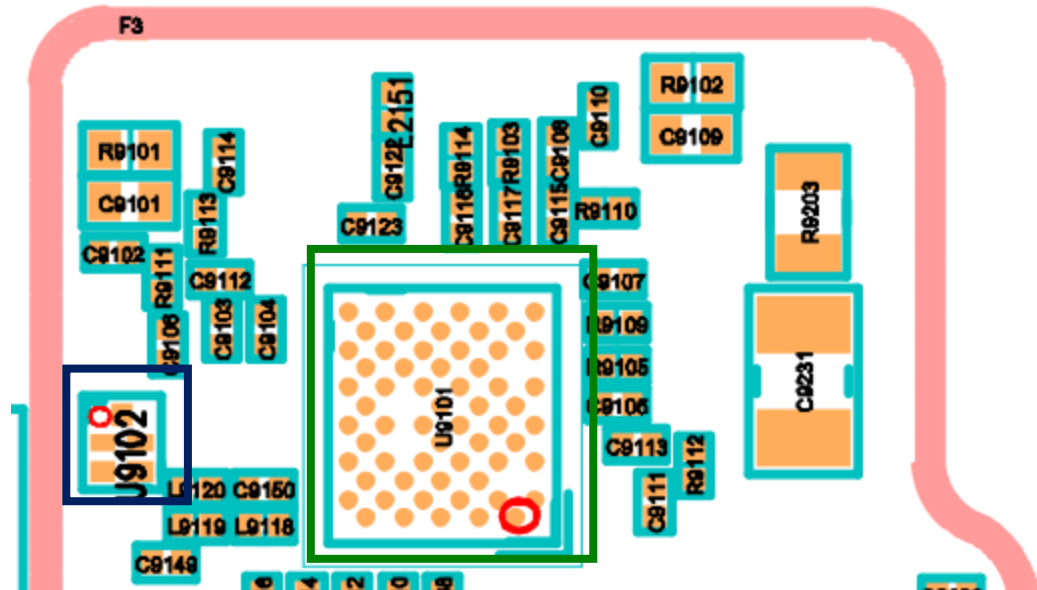
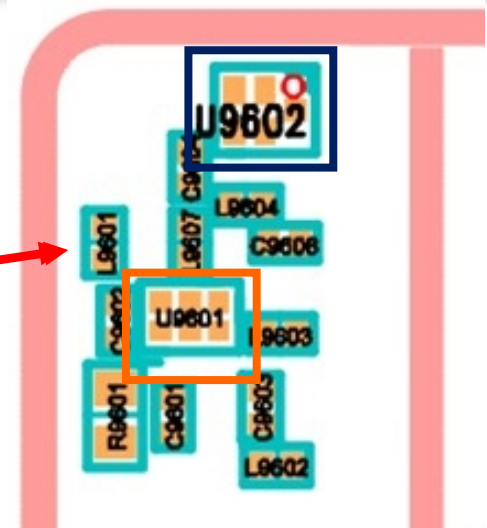


GPS Block Diagram



GPS Power Line

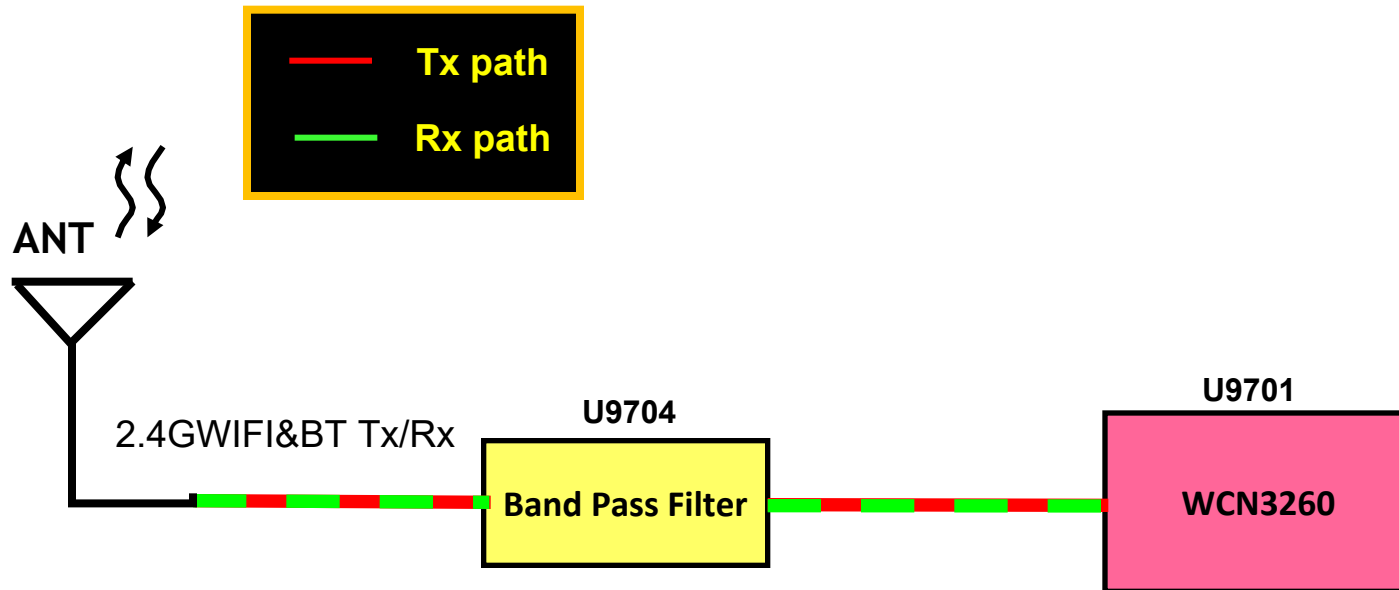
- SAW (SAFRN1G56AA0F0A) U9602,U9102
 - LNA (BGA824N6E6327) U9601
 - RFIC (WTR2606) U9101



GPS Function Test Procedure

- Check GPS function with GPS Test Tool.
- Check the C/N value with different signal level @ -80 dBm & -120 dBm.
- Check the GPS feed point.
- Check BGA824N6E6327 (U9601) and surrounding components
- Check the 1.8V supply voltage of GPS ENABLE PIN & 2.85V for LNA on.

WIFI/BT Block Diagram

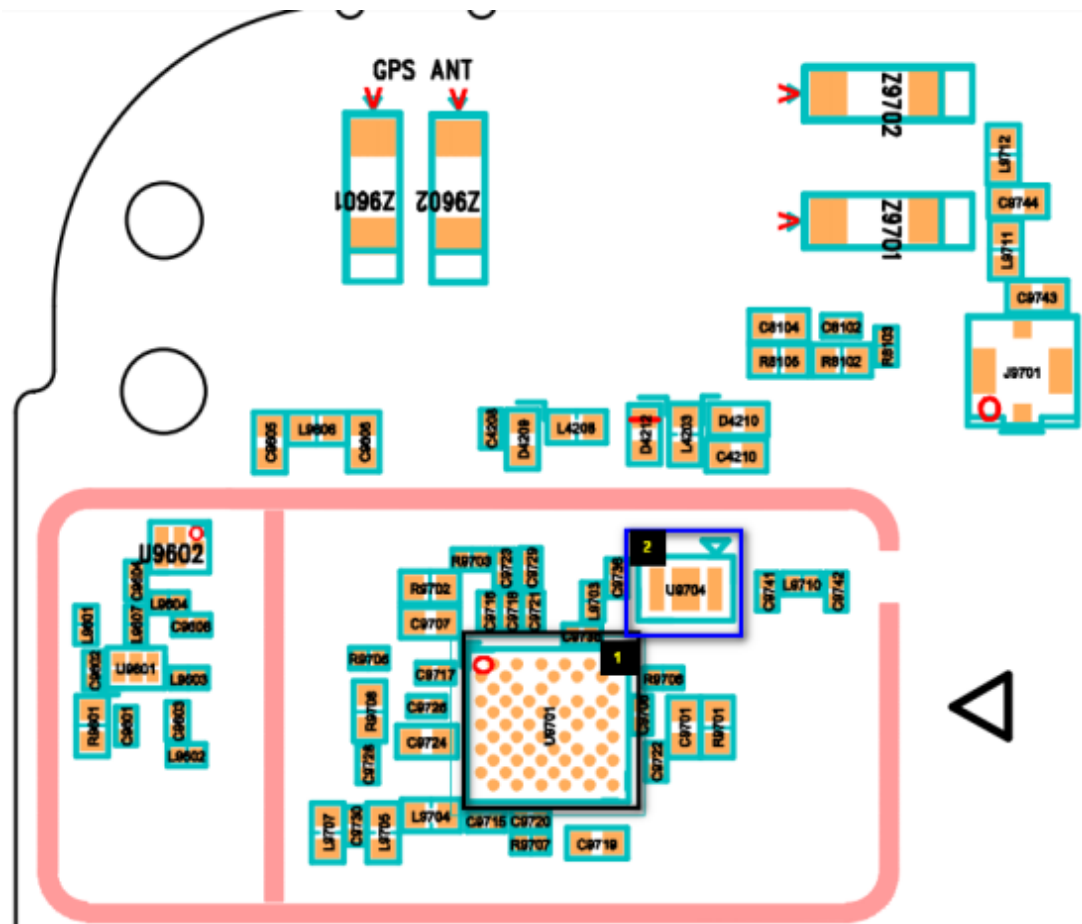


WIFI/BT 2.4G

RF Layout on Bottom

-  Rx Serial IC.
-  T/Rx Serial comp.
-  Serial IC.

- 1. U9701:** WIFI/BT Module
WCN3620
- 2. U9704:** WIFI/BT 2.4G
Band Pass Filter



BT/WLAN Function Test Procedure

- **Change antenna**
- **Check the BT/WLAN spring**
- **Check WCN3620 and surrounding components**

Thanks for Your attention !